Phenotype–Genotype Integration Using *Jedae-unshiu* Reference SNPs for Citrus Line Characterization

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Introduction

Citrus unshiu is a major fruit crop in Korea, cultivated primarily in the Jeju region, and breeding limitations such as seedlessness and high heterozygosity have led to the use of γ-irradiation–based mutation breeding. Among the resulting mutants, *Jedae-unshiu* exhibits distinctive peel traits, higher firmness, and elevated flavonoid levels, supported by SNP/InDel and transcriptomic evidence.

Building on these findings, this study integrates phenotypic patterns with SNP-based genetic information to characterize citrus line relationships using *Jedae-unshiu* as the reference cultivar.

Datasets

1. Genotype Data (WGS-based SNP Matrix)

Whole-genome sequencing of seven citrus lines yielded a high-quality set of approximately 660,000 SNPs. Among these, 160 markers specific to *Jedae-unshiu* were identified, with each SNP containing IUPAC-encoded genotypes and read-depth information. IBS genetic distances were then calculated across the seven lines using *Jedae-unshiu* as the reference cultivar.

2. Phenotype Data (2022–2024 Fruit Measurements)

Fruit traits from seven citrus lines (including *Jedae-unshiu*) were collected from 2022 to 2024, covering size, color, peel thickness, sugar content, and acidity. To address annual variability, year-effect normalization was applied. The processed dataset was then used for PCA, clustering, and trait profiling.

Phenotype Overview

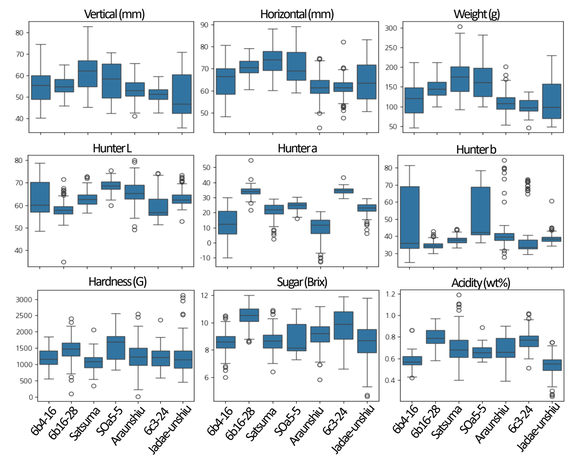

Figure 1. Phenotype Distributions Across Citrus Lines

Most traits showed clear line-to-line differences, especially in size, weight, and peel-related characteristics.

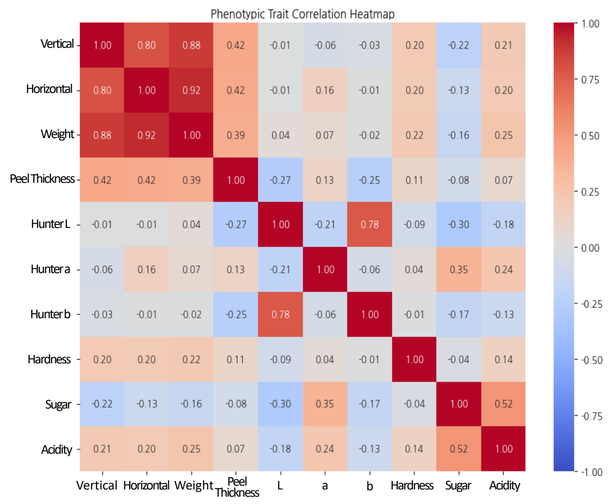

Figure 2. Trait Correlation Structure

Size, weight, and peel thickness were strongly correlated, while Brix and acidity showed a positive association; a* had a weak link to sugar content.

Method

Data Preprocessing → ***Phenotype Analysis*** → ***Genotype Analysis*** → ***Integrated Interpretation***

1. Data Preprocessing

Phenotype data were normalized to remove year effects, and all traits were scaled to ensure comparability across lines. For genotype analysis, a pre-filtered high-quality SNP set from seven citrus lines was used after aligning the matrix to shared loci.

2. Phenotype Analysis

PCA and clustering (hierarchical and k-means) were performed to identify line-specific phenotypic patterns, and trait loading vectors were examined to determine which traits contributed most to cluster separation.

3. Genotype Analysis

Provided SNPs were encoded using IUPAC genotype codes, which represent heterozygous states with single-letter ambiguity symbols. IBS genetic distances were calculated using *Jedae-unshiu* as the reference cultivar.

4. Integrated Interpretation

Phenotype-based cluster patterns were compared with IBS similarity structure. Variation patterns associated with trait differences and *Jedae-unshiu*–like lines were evaluated.

Results

1. Data Preprocessing

PCA revealed separation by year in phenotype data, indicatingannual effects.

After normalization, line-specific patterns became more comparable across years.

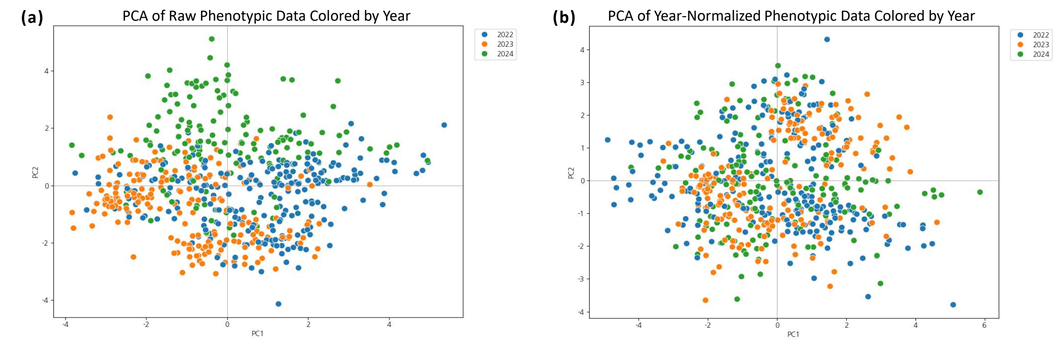

Figure 3. PCA comparison before and after year-effect normalization.

2. Phenotype Analysis

Phenotypic analysis revealed three major trait-driven clusters, with partial separation in PCA and clearer structure through K-means clustering.

The cluster composition and hierarchical grouping placed *Jedae-unshiu* near *6b4-16* and *Araunshiu*, indicating shared phenotypic characteristics.

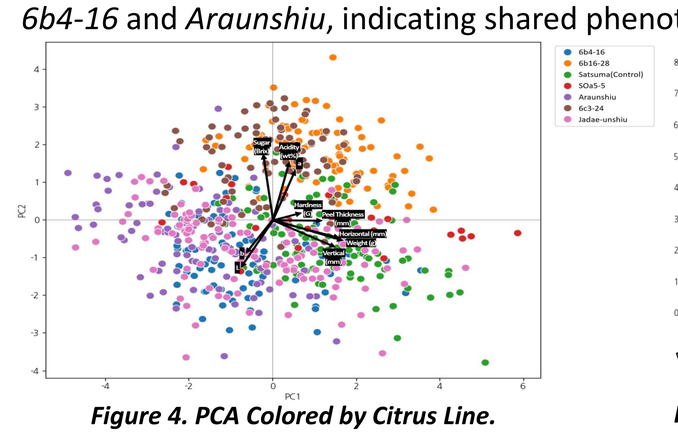

Figure 4. PCA Colored by Citrus Line.

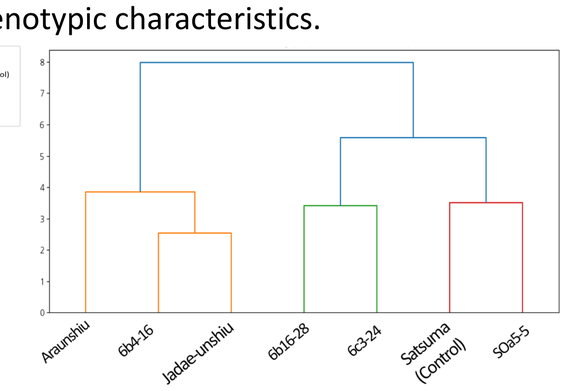

Figure 5. Hierarchical Clustering of Citrus Lines.

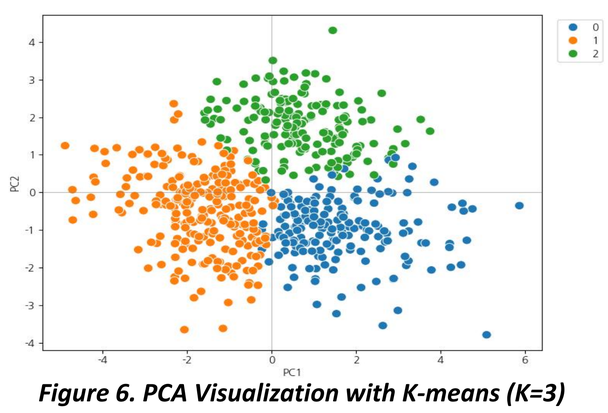

Figure 6. PCA Visualization with K-means (K=3) Cluster Assignments.

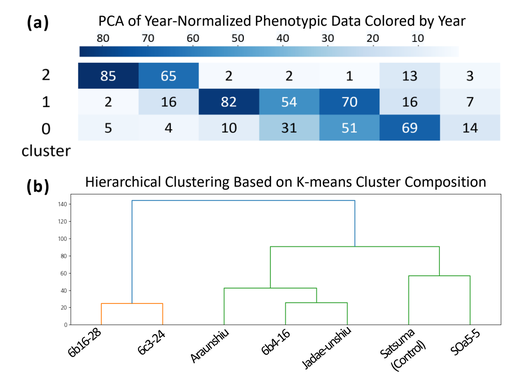

Figure 7. K-means cluster composition and cluster-based hierarchical grouping of citrus lines.

3. Genotype Analysis

Genotype analysis using IBS distance revealed a broadly similar level of genetic variation across lines, yet *Araunshiu*, *satsuma* and *6b4-16* showed the shortest distances to *Jedae-unshiu*. Importantly, this proximity pattern aligns with phenotype-based clustering, suggesting that the observed trait similarities are supported by underlying genetic structure.

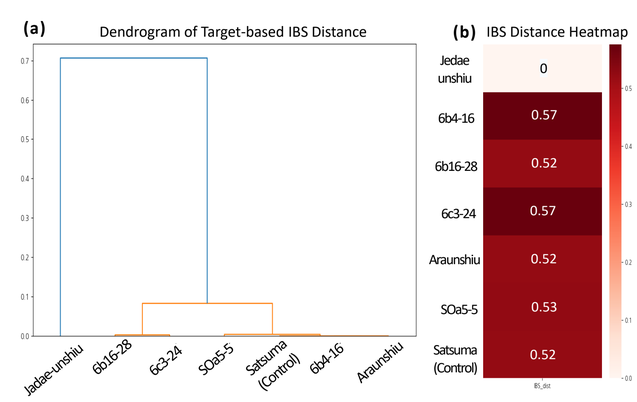

Figure 8. Genotype analysis results based on IBS distance using Jedae-unshiu as the reference.

4. Integrated Interpretation

PCA loadings show that hardness, color traits (L, a, b), and acidity were major factors driving the cluster including *Jedae-unshiu*, *Araunshiu*, and *6b4-16*, with sugar contributing weakly. When compared with genotype-based IBS distances, this phenotype-defined cluster—together with *Satsuma*—also showed the closest genetic proximity to Jedae-unshiu. The consistency between phenotypic clustering and genomic similarity suggests that some of the 160 *Jedae-unshiu*–specific SNP markers may be associated with traits such as acidity, sugar content, and color characteristics.

Conclusions

Phenotypic variation and IBS-based genetic similarity presented a broadly consistent relationship among citrus lines. This highlights the value of data-driven integration for revealing line-level structure and supporting marker-informed citrus breeding.